

STAGE 6: MISSING CONTEXT DATASET ANALYSIS

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Introduction

This report studies a daily dataset from 2018 to 2024 that shows clear rhythms, sudden breaks, and shifting relationships among five variables. The goal is to understand what kind of system this might represent, why its behavior changes over time, and what the major anomalies reveal. The patterns suggest seasons, pressure, adaptation, and recovery, but the true context is unknown.

Data Ontology

The dataset comprises daily observations from 2018 to 2024 of five unnamed numerical variables: Var1 to Var5 with date being known. Initial statistical exploration reveals:

- Var1: A high-magnitude activity metrics of 7 to 156. Before 2020, it shows a strong seasonal peak cycle of each year. This looks like physical movement or an economic throughput.
- Var2: is centered near zero from -0.7 to 0.7. This looks like a confidence level or sentiment.
- Var3: A low-magnitude counterpart to Var1 of 0 to 22. It mirrors Var1 closely with the cycle. It looks like local transactions or micro interactions.
- Var4: Sparse binary spikes of 0 or 1. It is uncorrelated with other variables. which might be a trigger or an event flag such as holidays, policy changes, or system interventions.
- Var5: is a magnitude from 2 to 60 that behaves inversely to Var1 and Var3. This might be a substitution signal or a digital proxy like remote work logins, home internet usage, or digital engagements.

The correlation analysis confirms this structure as shown in figure 1:

- Var1 and Var3: $r = +0.76$ suggesting tightly coupled. These two variables are linked conventionally.
- Var1 and Var5: $r = -0.82$ suggesting strong substitution effect implying causality.
- Var2 moderately correlates with Var1: $r = +0.42$ suggesting sentiment influences behavior. This also implies causality
- Var4 is uncorrelated, suggesting an external trigger. Which confirms conventional

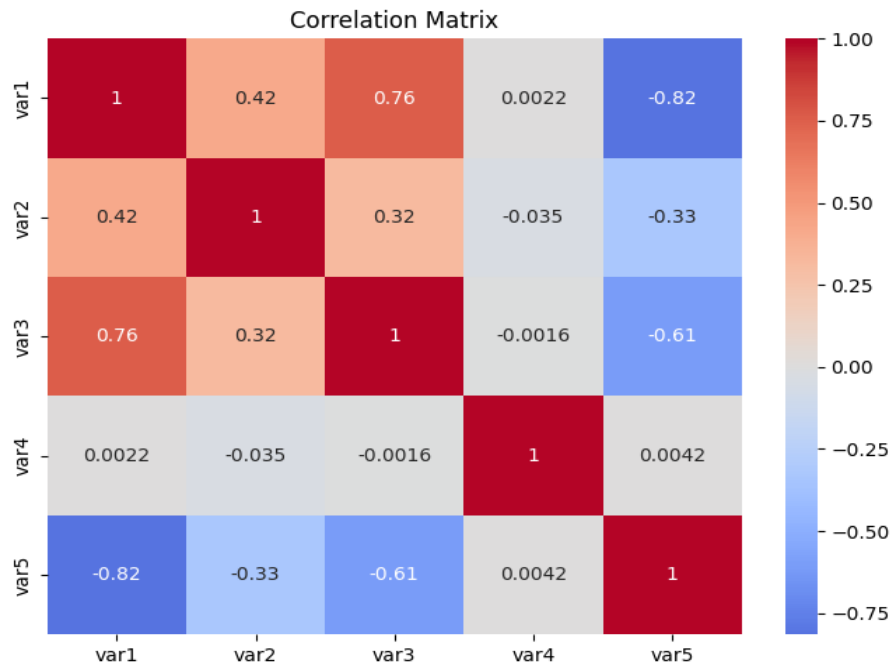


Fig 2; the heatmap plot showing the correlations among the variable.

The data exhibits seasonal cycles, sharp drop, lowered activity in response to recovery which never returned to the seasonal cycles. With this, it is most likely that this dataset reflects a human behavioral system, something like a city economy or a social environment reacting to a major shock.

Hypothesis Taxonomy

I identified three structural anomalies using rolling z-score analysis and the multivariate Anomaly Detection. The following are the three anomalies detected:

First Anomaly

When did it happen: August 2019. I observed that Var3 dropped from 16.2 to 10 with Var1 from 136 to 82 while other variables are stable. Therefore, my hypotheses are:

“Local supply chain shock such as port strike or small business regulation, Social media panic affecting niche activity and Isolated sensor error or data pipeline glitch”

Plausibility Ranking: The economic explanation is most convincing because the drop resembles a local shock, the behavioral explanation is possible but less stable for a single day, and the technical explanation is least likely since a system glitch seems improbable.

Second Anomaly

When did it happen: March 2020.

My observation shows that Var3 dropped from 15 to 5 and Var1 dropped from 100 to 20 while Var5 surges higher from 30 to 45. Therefore, my hypotheses are:

“Global lockdowns due to pandemic which reduced physical activity and caused increased in digital surges, Mass panic triggers voluntary withdrawal from public spaces, and natural disasters like earthquakes and floods that force evacuation”

Plausibility Ranking: The economic explanation is strongest because it fits the global timeline and the uneven recovery; though the behavioral explanation is possible but less precise and doesn't match policy-level timing; and the environmental explanation is least likely since natural disasters don't produce a sustained digital shift or the recovery pattern seen here.

Third Anomaly

When did it happen: May 2021.

I noticed that Var3 surges from 5 to 15.9, Var 1 rise along with Var3 from 14 to 137. Var 5 on the other end reduced drastically from 54 to 25. Therefore, my hypotheses are:

“Vaccination rollout and stimulus spending enable reopening, Social reintegration after prolonged isolation, and Data normalization reset or reporting change”

Plausibility Ranking: The economic explanation is most convincing because the recovery is gradual rather than abrupt; the behavioral explanation is possible but not strong enough to explain the scale; and the technical explanation is least likely since the smooth trends before and after contradicting a system glitch.

First Narrative

Assuming the dataset is from a global city, recording the activities like the following.

Var1: Daily foot traffic in commercial districts.

Var2: Investors or Business Owner confidence level.

Var3: Small business and startup sales performance.

Var4: Trade restriction.

Var5: Home internet usage or remote work logins.

The system exhibited a consistent seasonal cycle until August 2019, when an anomaly disrupted this pattern, likely due to regional economic shocks from US-China trade tensions (Stankiewicz, 2019). This disruption is evidenced by a decline in Var3 followed by Var2 turning negative, indicating reduce in business owners' and investors' confidence as stated by Brian Riley in the CNBC news report in September 2019. Another significant anomaly occurred in March 2020 with the onset of a global health crisis, leading to lockdown that halted business activity of (Var3) (Koster et al., 2022) and caused a sharp drop in foot traffic (Var1), while remote work and internet usage (Var5) surged higher (Silver, 2023). Although the city began a cautious recovery by May 2021, foot traffic gradually improved and remote work remained elevated compared to pre-pandemic levels, suggesting a lasting change in work and economic patterns. These observations highlight how external shocks can cause structural disruptions in the system, but adaptation mechanisms have enabled a form of resilience and survival in the post-pandemic period.

The Counter Narrative

What if the data assumed is not recorded from human or natural activities rather it is a telemetry from a multi-agent simulation like a prototype for a smart city or a digital twin environment. The dataset interpretation below is assumed to be from remote monitoring.

Var1: simulated vehicle traffic.

Var2: system health or integrity score.

Var3: Non-player character NPC interactions such as simulated small business activity.

Var4: scheduled maintenance triggers.

Var5: physically controlled vehicle traffic.

For this interpretation, August 2019 anomaly is not from regional economic shock but rather the perceived negative effect in system health (Var2) is a simulation bug or rule update that reduced NPC interactions (Var3) which indicate the drop in simulated small business activity like patterns common in video games where glitches disrupt behavioral coherence (*Stathis, 2020*). Then in March 2020, developers intentionally activated a global crisis module from a stress-test scenario modeling a pandemic. This module forced a halt on simulated physical movement indicating the crash of simulated vehicle (Var1) and the non-player character showing why Var3 collapses (*Wang et al., 2024*) while redirecting activity to physical human-controlled vehicle (Var5) making it surge high. The synchronization is not coincidence, rather it is a scripted behavior. In May 2021 the anomaly recovery was not organic, it was a rollback or fix that re-enabled movement and NPC interactions. Yet, the system health score Var2 never fully recovered because of error log or residual anomalies remained in the simulation state. Critically, this framework exposes developer intent as the core driver, questioning real-world analogies and highlighting simulation abstraction limits.

Reflection on Epistemic Uncertainty

Since we are analyzing data without context, uncertainty is built into every interpretation.

Several assumptions hold the narrative together:

- I assumed it is daily data from different systems, but it could be the same system.
- I suggested human behavior due to sentiment influenced patterns, but it could be from natural factors and AI systems can as well mimic emotion.
- I referenced the 2020 pandemic it could be engineered logic or a natural disaster.

My narrative will be false if:

1. Var5 is revealed to be temperature in Celsius indicating natural factor system
2. Var4 is proven to be random noise that was not recorded from event.
3. External data shows no global crisis in 2020 but a video game simulation.

Quantifying my Epistemic Uncertainty

The chart below visualizes the epistemic statistical uncertainty computing a 60-day rolling mean and standard deviation.

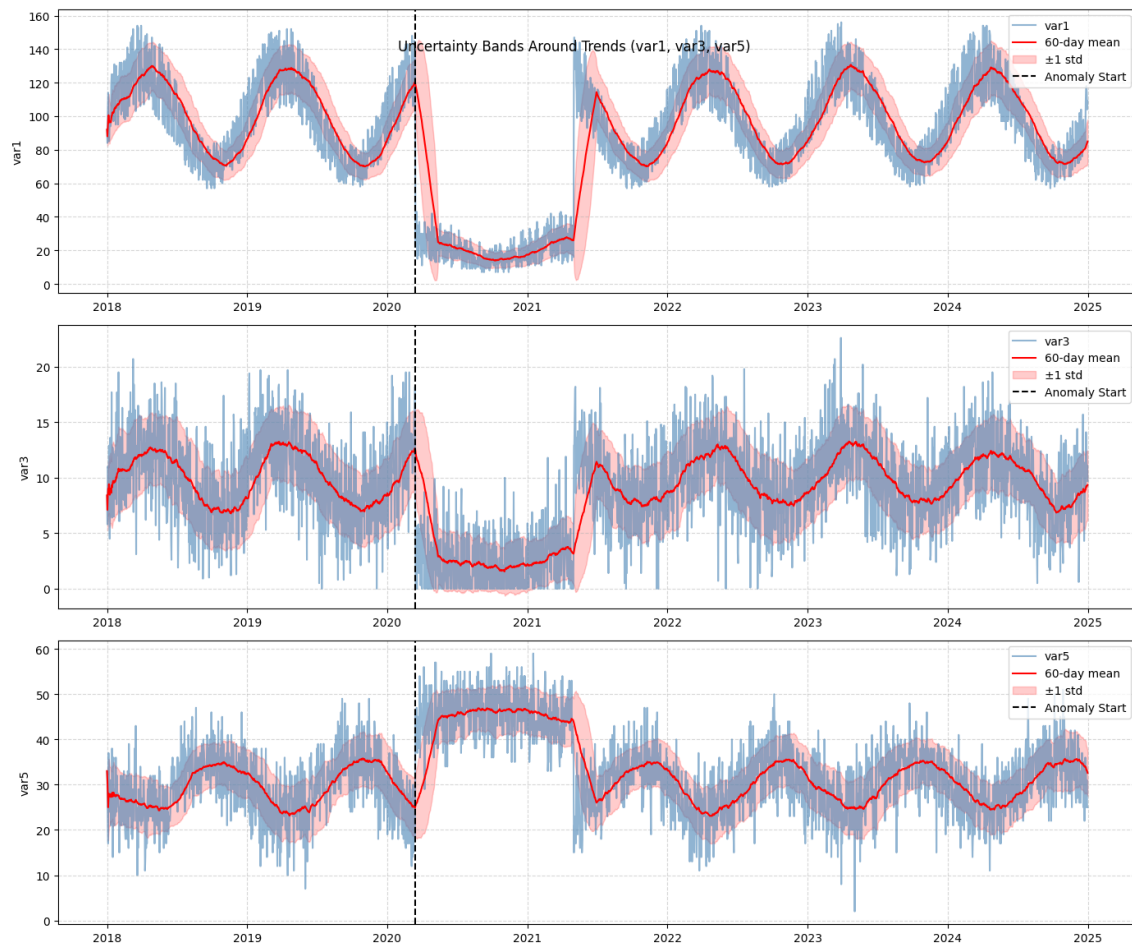


Fig 2: charts to quantify Uncertainty.

Key Insight and Findings

Var1 falls sharply from 100 to 20 and also Var3 shows a similar collapse from 15 to 5, more than three standard deviations below its trend. At the same time, Var5 rises from 30 to 45, placing it about two standard deviations above its normal range. So, quantifying my epistemic uncertainty.

- Statistically: in March 2020, Var1 and Var3 drop exceed the 3std implying that has 0.3% probability of happening under normal conditions.
- Interpretively: I estimate 70% confidence that this is a human behavioral system, 20% that it is a simulation and 10% that it is something else.
- Even if I am sure that my hypotheses are anomaly the cause is still unknown.

Conclusion

The analysis shows a system that moves steadily until three disruptions reshape its direction. The strongest explanation points to economic or behavioral shocks, especially around 2020, though other interpretations remain possible. The system adapts, settles, and forms a new pattern that never fully returns to its earlier state.

References

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